
Department of Civil Engineering, IIT Madras
CV2090 (Probability & Statistics for Civil Engineers)
Jul–Nov 2026

Practice Problem Set

Oct, 2026

Multivariate Random Variables & Joint Distributions

Q1. Joint Discrete Distributions and Correlation Two discrete random variables X and Y representing structural crack counts at two different critical bridge joints have the following joint probability mass function $p_{XY}(x, y)$:

$p_{XY}(x, y)$	$y = 0$	$y = 1$
$x = 0$	3/10	2/10
$x = 1$	1/10	4/10

- (a) Find the marginal PMFs $p_X(x)$ and $p_Y(y)$.
- (b) Determine whether X and Y are statistically independent.
- (c) Compute $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\text{COV}(X, Y)$, and the correlation coefficient ρ_{XY} . ■

Q2. Independent Continuous Variables and Joint CDFs Let X and Y be independent continuous random variables representing non-dimensionalized structural dead load and live load, with marginal CDFs $F_X(x) = 1 - e^{-2x}$ for $x \geq 0$ and $F_Y(y) = y^2$ for $0 \leq y \leq 1$.

- (a) Write the joint CDF expression $F_{XY}(x, y)$.
- (b) Derive the joint probability density function $f_{XY}(x, y)$ using partial differentiation.
- (c) Compute the probability $P(X \leq 0.5, Y \leq 0.5)$. ■

Q3. Linear Combination of Correlated Random Variables A column supports a dead load D and a live load L . Suppose $\mathbb{E}[D] = 120$ kN, $\mathbb{E}[L] = 80$ kN, $\sigma_D = 12$ kN, $\sigma_L = 16$ kN, and the correlation coefficient is $\rho_{DL} = 0.5$. The total load is $T = 1.2D + 1.6L$.

- (a) Calculate the mean total load μ_T .
- (b) Compute the covariance $\text{COV}(D, L)$ and the variance $\text{Var}(T)$.
- (c) Assuming T is normally distributed, evaluate the probability $P(T > 300 \text{ kN})$. ■

Q4. Bivariate Normal Distribution and Conditional Properties Let X and Y follow a bivariate normal distribution with $\mu_X = 10$, $\mu_Y = 20$, $\sigma_X = 2$, $\sigma_Y = 4$, and $\rho = 0.6$.

- (a) Write the expression for the conditional PDF $f_{Y|X}(y|x)$.
- (b) Determine the conditional mean $\mathbb{E}[Y | X = 12]$ and conditional variance $\text{Var}(Y | X = 12)$.
- (c) Evaluate $P(Y > 22 | X = 12)$ using standard normal distribution tables. ■

Q5. Joint PDF over Triangular Wedge Domain Consider two continuous random variables X (capacity) and Y (demand) with joint PDF defined over a triangular domain:

$$f_{XY}(x, y) = c e^{-x}, \quad 0 < y < x < \infty$$

- (a) Determine the normalization constant c .
- (b) Derive the marginal PDF $f_X(x)$.
- (c) Find the conditional density $f_{Y|X}(y|x)$ and evaluate $\mathbb{E}[Y | X = x]$. ■

Transformations of Random Variables

Q6. Monotonic Transformation of Continuous Variable Let X be an exponential random variable with parameter $\lambda > 0$, having PDF $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Consider the transformation $Y = e^X$.

- (a) Determine the support domain of Y .
- (b) Derive the PDF $f_Y(y)$ using the change-of-variable (Jacobian) technique.
- (c) Identify the distribution family of Y . ■

Q7. Symmetric Non-Monotonic Transformation Let $X \sim U(-2, 2)$ be a uniform random variable over the interval $[-2, 2]$. Let $Y = X^2$.

- (a) Find the CDF $F_Y(y)$ for $0 \leq y \leq 4$.
- (b) Differentiate $F_Y(y)$ to obtain the PDF $f_Y(y)$.
- (c) Calculate the expected value $\mathbb{E}[Y]$ directly from $f_Y(y)$. ■

Q8. Linear Transformation of Normal Vectors Let $X_1 \sim N(5, 4)$ and $X_2 \sim N(10, 9)$ be independent normal random variables. Define $Y = 2X_1 - X_2 + 3$.

- (a) Identify the distribution of Y .
- (b) Compute the mean μ_Y and variance σ_Y^2 .
- (c) Calculate $P(Y \leq 0)$. ■

Q9. Distribution of Structural Stress Ratio Let $Y = X^3$, where X is a continuous variable with PDF $f_X(x) = 2x$ for $0 \leq x \leq 1$.

- (a) Determine the cumulative distribution function $F_X(x)$.
- (b) Derive the CDF $F_Y(y)$ for $0 \leq y \leq 1$.
- (c) Compute the PDF $f_Y(y)$. ■

Extreme Value Distributions & Tail Behavior

Q10. Exact and Asymptotic Maxima for Exponential Parent Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) random variables with an exponential parent PDF $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Let $Y_n = \max(X_1, X_2, \dots, X_n)$.

- (a) Write the exact CDF $F_{Y_n}(y)$ for a sample of size n .
- (b) Show that as $n \rightarrow \infty$, the exact CDF $F_{Y_n}(y)$ can be expressed in the Gumbel Type I asymptotic form:

$$F_{Y_n}(y) \approx \exp[-\exp(-\alpha_n(y - u_n))]$$

by proving that $u_n = \frac{\ln n}{\lambda}$ and $\alpha_n = \lambda$.

- (c) Highlight that in this limiting Gumbel distribution, u_n is called the **location parameter** (representing the mode or characteristic largest value) and α_n is called the **scale parameter** (representing an inverse measure of dispersion).
- (d) Evaluate the numerical values of the location parameter u_n and scale parameter α_n for $\lambda = 0.5$ and sample size $n = 100$. ■

Q11. Exact and Asymptotic Minima for Weakest-Link Modeling Let $X_1, X_2, \dots, X_n$ represent compressive strengths of structural fibers with parent CDF $F_X(x) = \left(\frac{x}{w}\right)^k$ for $0 \leq x \leq w$.

-
- (a) Write the exact survival probability $P(Y_1 > y)$ (i.e., $1 - F_{Y_1}(y)$) for the sample minimum $Y_1 = \min(X_1, \dots, X_n)$.
 - (b) Show that as $n \rightarrow \infty$, the limiting distribution of Y_1 converges to a Weibull Type III distribution.
 - (c) Explain why this distribution models the failure strength of chains or brittle concrete elements. ■

Q12. Extreme Rainfall Return Periods and Overtopping Risk Annual maximum 24-hour rainfall Y at a reservoir station follows a Gumbel Type I distribution with location parameter $u = 80$ mm and scale parameter $\alpha = 0.05$ mm⁻¹.

- (a) Calculate the 50-year design rainfall level y_{50} .
- (b) Calculate the 100-year design rainfall level y_{100} .
- (c) Evaluate the probability that the 100-year rainfall level will be exceeded at least once during a 30-year operational life. ■

Q13. Fréchet Distribution for Heavy-Tailed Extremes The annual maximum daily river discharge Y follows a Fréchet (Type II) distribution with CDF $G(y) = \exp \left[- \left(\frac{v_n}{y} \right)^k \right]$ for $y > 0$, where $v_n = 500$ m³/s and shape parameter $k = 3$.

- (a) Express the T -year return period discharge y_T in terms of T , v_n , and k .
- (b) Compute the 200-year flood discharge y_{200} .
- (c) Compare y_{200} with the characteristic value v_n and discuss the heavy-tail growth rate. ■

Q14. Asymptotic Behavior of Uniform Maxima Let $X_1, X_2, \dots, X_n$ be i.i.d. uniform random variables on $[0, \theta]$. Let $M_n = \max(X_1, \dots, X_n)$.

- (a) Derive the exact CDF $F_{M_n}(m)$ for $0 \leq m \leq \theta$.
- (b) Derive the PDF $f_{M_n}(m)$.
- (c) Calculate the mean $\mathbb{E}[M_n]$ and show that $M_n \rightarrow \theta$ as $n \rightarrow \infty$. ■

Parameter Estimation & Confidence Intervals

Q15. Method of Moments Estimation (MoM) The probability density function (PDF) of a Rayleigh distributed random variable X with scale parameter $\alpha > 0$ is given by:

$$f_X(x; \alpha) = \frac{x}{\alpha^2} \exp \left(-\frac{x^2}{2\alpha^2} \right), \quad x \geq 0$$

The theoretical mean and standard deviation of the Rayleigh distribution in terms of α are:

$$\mathbb{E}[X] = \alpha \sqrt{\frac{\pi}{2}}, \quad \sigma_X = \alpha \sqrt{2 - \frac{\pi}{2}}$$

A total of $n = 10$ measurements of maximum wind speeds (m/s) during storm events were recorded as follows:

12.1, 14.3, 10.8, 15.2, 11.5, 13.0, 12.7, 14.0, 11.8, 13.6

- (a) Calculate the sample mean $\bar{x}$ and sample standard deviation s for the dataset.
- (b) Estimate the scale parameter α using the MoM by equating $\bar{x}$ to $\mathbb{E}[X]$.
- (c) Using $\hat{\alpha}_{\text{MoM}}$, compute the theoretical standard deviation σ_X of the fitted Rayleigh model, compare it with s , and explain the discrepancy. ■

Q16. Maximum Likelihood Estimation for Gamma Scale Parameter Let $X_1, X_2, \dots, X_n$ be an i.i.d. sample from a Gamma distribution with known shape parameter $k = 2$ and unknown scale parameter $\beta > 0$, having PDF $f_X(x; \beta) = \beta^2 x e^{-\beta x}$ for $x \geq 0$.

- (a) Write the joint likelihood function $L(\beta)$.
- (b) Formulate the log-likelihood function $\ln L(\beta)$ and evaluate $\frac{d \ln L}{d \beta} = 0$.
- (c) Derive the explicit formula for the maximum likelihood estimator $\hat{\beta}_{\text{MLE}}$. ■

Q17. Unbiasedness Properties of Sample Variance Estimators Let $X_1, X_2, \dots, X_n$ be an i.i.d. random sample with mean μ and variance σ^2 . Consider $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ and $S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$.

- (a) Prove that $\mathbb{E}[S^2] = \sigma^2$.
- (b) Calculate the bias $B(S_n^2) = \mathbb{E}[S_n^2] - \sigma^2$.
- (c) Explain why S^2 is preferred over S_n^2 for small sample sizes n . ■

Q18. Two-Sided and One-Sided Confidence Intervals for Normal Mean A batch of $n = 12$ steel rebar specimens was tested for yield strength (MPa). The calculated sample mean is $\bar{x} = 460$ MPa. Assume the strength is normally distributed with a known standard deviation $\sigma = 15$ MPa.

- (a) Construct a 95% two-sided confidence interval for the population mean μ .
- (b) Construct a 95% lower confidence bound for μ .
- (c) Calculate the sample size n required to restrict the 95% two-sided margin of error to at most 5 MPa. ■

Q19. Small-Sample Confidence Interval and Chi-Square Variance Bounds Seven concrete core cylinders were tested for elastic modulus (GPa), yielding sample mean $\bar{x} = 28.5$ GPa and sample standard deviation $s = 2.4$ GPa. Assume the population is normally distributed.

- (a) Construct a 95% two-sided confidence interval for the mean μ using Student's t -distribution.
- (b) Construct a 95% two-sided confidence interval for the population variance σ^2 using the Chi-Square distribution ($\chi_{0.975,6}^2 = 1.237$, $\chi_{0.025,6}^2 = 14.449$).
- (c) Deduce the corresponding 95% confidence interval for the population standard deviation σ . ■

Standard Normal Probabilities

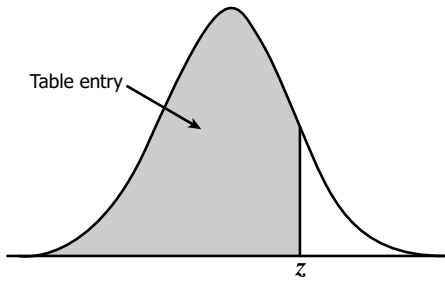


Table entry for z is the area under the standard normal curve to the left of z .

[illegible]



t Table

cum. Prob	$t_{1-\alpha, df}$										$Z_{1-\alpha}$
one-tail (α)	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level (two-tailed)										

Note: For df larger than 100, use the 1000 value; for larger than 1000, use the Z value